

COSMOS Cluster 12

Machine Learning

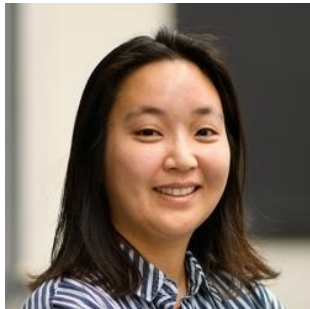
Summer 2026

Topic: Natural Language Processing

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Teacher Fellow: Emmanuel Pizano

UC San Diego

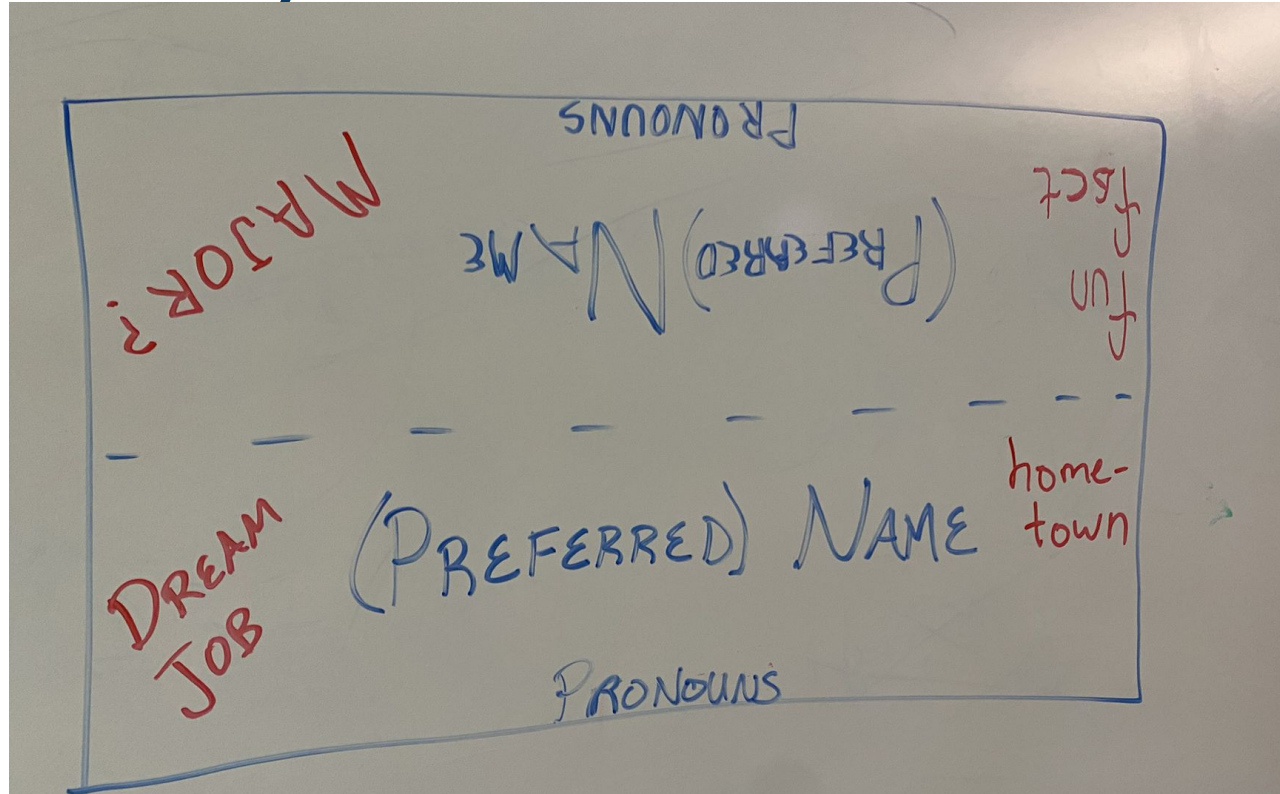


Sooh Nam Liao

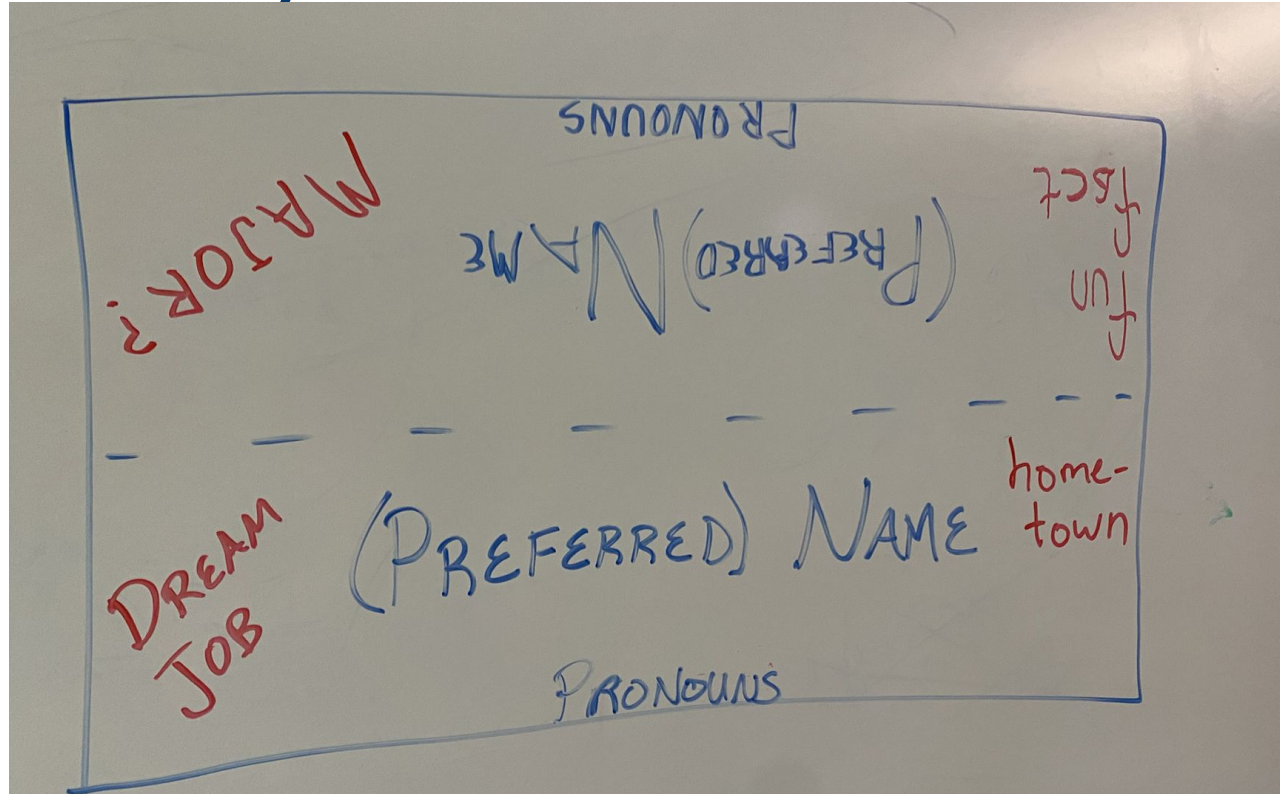
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- Research Interests:
 - Promoting healthy study strategies at scale by
 - using AI-powered educational tools for better student metacognition & instructor/student interaction
 - psychotherapy techniques on a mobile application (willowmind.io)
- Teaching:
 - Data structures and algorithms
 - Introduction to computer science
- Fun: hiking/camping, k-dramas, parenting two kids (7 and 3)

Let's create the name tags so I can remember your name



Let's create the name tags so I can remember your name



Natural Language Processing

Discuss:

1. What do you think Natural Language Processing (NLP) is?
2. What are some applications of NLP?

AI works with many different types of data

- tables
- images
- sounds
- texts

NLP deals with text data

1. text analysis

- a. analyze given text and derive any insight

2. text generation

- a. auto-completion, auto-correction

3. And we are going to work on a simplest NLP application, called sentiment analysis

Sentiment analysis determines whether a given text is positive or negative

Given a movie review,

“it was a terrific movie” -> positive

“it was an awful movie” -> negative

The analysis follows a machine learning model, called Naive Bayes classifier

Naive Bayes Classifier

Following is a dataset of 1000 fruits. The dataset consists of four columns. The first column (Fruit Name) has three unique values - Orange, Banana, and Other. The rest three columns are binary data.

Fruit	Long	Sweet	Yellow
Orange	0	1	0
Banana	1	0	1
Banana	1	1	1
Other	1	1	0
..	..	..	..

Answer:

What is $P(\text{Fruit is a Banana})$?

What is $P(\text{Fruit is a Long Banana})$?

What is $P(\text{Fruit} = \text{Banana} \mid \text{Long} = 1)$?

What is $P(\text{Sweet} = 1 \mid \text{Fruit} = \text{Banana})$?

Summary of the raw dataset

Fruit	Long	Not Long	Sweet	Not Sweet	Yellow	Not Yellow	Total
Banana	400	100	350	150	450	50	500
Orange	0	300	150	150	300	0	300
Other	100	100	150	50	50	150	200
Total	500	500	650	350	800	200	1000

You are given a mystery fruit and want to figure out what it is, using the same dataset.

What would you do?

You are given a mystery fruit and want to figure out what it is, using the same dataset.

In ML/AI terms:

You can measure information on the three features of the unknown data sample: x_1 (Long?), x_2 (Sweet?), and x_3 (Yellow?).

Then you want to classify y (Fruit Yype), given all the feature measures.

Question: Write what kind of probability we are calculating.

Naive Bayes classifier is based on “Bayes rule”

and Bayes rule is all about conditional probability!

$$P(Y|X) = \frac{P(X|Y) * P(Y)}{P(X)}$$

And there are names for three terms

$$\text{Posterior Probability } P(Y|X) = \frac{\text{Likelihood of Evidence } P(X|Y) * \text{Prior Probability } P(Y)}{P(X)}$$

Following is a dataset of 1000 fruits. The dataset consists of four columns. The first column (Fruit Name) has three unique values - Orange, Banana, and Other. The rest three columns are binary data.

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Other	100	100	150	50	50	150	200
Total	500	500	650	350	800	200	1000

Question:

You are given a fruit that is long, sweet, and yellow. What is that fruit the most likely to be?

Answer: The fruit which has the maximum of following probabilities!

$P(\text{Fruit} = \text{Banana} \mid (\text{Long}=1, \text{Sweet}=1, \text{and Yellow}=1))$

$P(\text{Fruit} = \text{Orange} \mid (\text{Long}=1, \text{Sweet}=1, \text{and Yellow}=1))$

$P(\text{Fruit} = \text{Other} \mid (\text{Long}=1, \text{Sweet}=1, \text{and Yellow}=1))$

Answer: The fruit which has the maximum of following probabilities!

$P(\text{Fruit} = \text{Banana} \mid (\text{Long}=1, \text{Sweet}=1, \text{and Yellow}=1))$

$P(\text{Fruit} = \text{Orange} \mid (\text{Long}=1, \text{Sweet}=1, \text{and Yellow}=1))$

$P(\text{Fruit} = \text{Other} \mid (\text{Long}=1, \text{Sweet}=1, \text{and Yellow}=1))$

$$P(Y|X) = \frac{P(X|Y) * P(Y)}{P(X)}$$

Y: Fruit = Banana

X:

1) Long = 1

2) Sweet = 1

3) Yellow = 1

“Naive” Bayes assumes no dependence across input features

1. Meaning there is no correlation between a fruit being long, sweet, and yellow (= they are all independent events)
2. This allows us to calculate a joint probability easily
 - a. $P(\text{Long and Banana}) = P(\text{Long}) * P(\text{Banana})$

Fruit	Long	Not Long	Sweet	Not Sweet	Yellow	Not Yellow	Total
Banana	400	100	350	150	450	50	500
Orange	0	300	150	150	300	0	300
Other	100	100	150	50	50	150	200
Total	500	500	650	350	800	200	1000

$P(\text{Fruit} = \text{Banana} \mid (\text{Long}=1 \text{ and Sweet}=1 \text{ and Yellow}=1))$

$$P(Y|X) = \frac{P(X|Y) \cdot P(Y)}{P(X)}$$

Y: Fruit = Banana

*X: Long = 1 **and** Sweet = 1 **and** Yellow = 1*

$$P(Y) = P(\text{Banana}) = 500/1000 = 0.5$$

Fruit	Long	Not Long	Sweet	Not Sweet	Yellow	Not Yellow	Total
Banana	400	100	350	150	450	50	500
Orange	0	300	150	150	300	0	300
Other	100	100	150	50	50	150	200
Total	500	500	650	350	800	200	1000

$P(\text{Fruit} = \text{Banana} \mid (\text{Long}=1 \text{ and Sweet}=1 \text{ and Yellow}=1))$

$$P(Y|X) = \frac{P(X|Y) * 0.5}{P(X)}$$

$Y: \text{Fruit} = \text{Banana}$

$X: \text{Long} = 1 \text{ and Sweet} = 1 \text{ and Yellow} = 1$

$$P(X|Y) = P(\text{Long} \mid \text{Banana}) * P(\text{Sweet} \mid \text{Banana}) * P(\text{Yellow} \mid \text{Banana})$$

$$P(\text{Long} \mid \text{Banana}) = 400/500 = 0.8$$

$$P(\text{Sweet} \mid \text{Banana}) = 350/500 = 0.7$$

$$P(\text{Yellow} \mid \text{Banana}) = 450/500 = 0.9$$

$$\begin{aligned} P(X|Y) \\ &= 0.8 * 0.7 * 0.9 \\ &= \mathbf{0.504} \end{aligned}$$

Fruit	Long	Not Long	Sweet	Not Sweet	Yellow	Not Yellow	Total
Banana	400	100	350	150	450	50	500
Orange	0	300	150	150	300	0	300
Other	100	100	150	50	50	150	200
Total	500	500	650	350	800	200	1000

$P(\text{Fruit} = \text{Banana} \mid (\text{Long}=1 \text{ and Sweet}=1 \text{ and Yellow}=1))$

$$P(Y|X) = \frac{0.504 * 0.5}{P(X)}$$

Y: Fruit = Banana

*X: Long = 1 **and** Sweet = 1 **and** Yellow = 1*

$$P(X) = P(\text{Long} \cap \text{Sweet} \cap \text{Yellow}) = P(\text{Long}) * P(\text{Sweet}) * P(\text{Yellow})$$

$$= 0.5 * 0.65 * 0.8 = 0.26$$

Fruit	Long	Not Long	Sweet	Not Sweet	Yellow	Not Yellow	Total
Banana	400	100	350	150	450	50	500
Orange	0	300	150	150	300	0	300
Other	100	100	150	50	50	150	200
Total	500	500	650	350	800	200	1000

$P(\text{Fruit} = \text{Banana} \mid (\text{Long}=1 \text{ and Sweet}=1 \text{ and Yellow}=1))$

$$P(Y|X) = \frac{0.504 * 0.5}{0.26} = 0.969$$

Y: Fruit = Banana

X: Long = 1 and Sweet = 1 and Yellow = 1

Fruit	Long	Not Long	Sweet	Not Sweet	Yellow	Not Yellow	Total
Banana	400	100	350	150	450	50	500
Orange	0	300	150	150	300	0	300
Other	100	100	150	50	50	150	200
Total	500	500	650	350	800	200	1000

$$P(\text{Banana} | X) = \frac{0.504 * 0.5}{0.26} = 0.969$$

When the fruit is long, sweet, and yellow, it's likely to be a banana!

$$P(\text{Orange} | X) = \frac{0 * 0.3}{0.26} = 0$$

$$P(\text{Other} | X) = \frac{0.0375 * 0.5}{0.26} = 0.072$$

Practical Consideration #1

$P(X)$ stays constant as 0.26, when calculating three different posterior probability

a. so we don't calculate it ..:D

$$P(Y|X) = \frac{P(X|Y) * P(Y)}{P(X)}$$

$$P(Banana|X) = \frac{0.504 * 0.5}{0.26} = 0.969$$

$$P(Orange|X) = \frac{0 * 0.3}{0.26} = 0$$

$$P(Other|X) = \frac{0.0375 * 0.5}{0.26} = 0.072$$

Practical Consideration #2

2. $P(\text{Orange}|X)$ becomes 0 simply because of a single term $P(\text{Long}|\text{Orange})$

- a. too extreme. so we lessen its impact with smoothing

$$P(Y|X) = \frac{P(X|Y) * P(Y)}{P(X)}$$

$$P(\text{Banana}|X) = \frac{0.504 * 0.5}{0.26} = 0.969$$

$$P(\text{Orange}|X) = \frac{0 * 0.3}{0.26} = 0$$

$$P(\text{Other}|X) = \frac{0.0375 * 0.5}{0.26} = 0.072$$

Laplace smoothing mitigates the impact of a single 0 term

To each $p(X|Y)$, add 1 to denominator, and # of input features to the numerator, when calculating the likelihood of evidence

$$P(Y|X) = \frac{P(X|Y) * P(Y)}{P(X)}$$

$$P(X|Y) = P(\text{Long} | \text{Orange}) * P(\text{Sweet} | \text{Orange}) * P(\text{Yellow} | \text{Orange})$$
$$= (0+1)/(300+3) = 0.003$$

Naive Bayes Classifier with Text Data

Text	Sentiment
just plain boring	neg
entirely predictable and lacks energy	neg
no surprises and very few laughs	neg
very powerful	pos
the most fun film of the summer	pos

Predict the sentiment of a text T: **“predictable with no fun”**

$$P(Y|X) = \frac{P(X|Y) * P(Y)}{P(X)}$$

Y: Sentiment = neg or pos

*X: each word in T (predictable, with, no, fun)
and we assume each word occurs
independently*

Text	Sentiment
just plain boring	neg
entirely predictable and lacks energy	neg
no surprises and very few laughs	neg
very powerful	pos
the most fun film of the summer	pos

Predict the sentiment of a text T:
“predictable with no fun”

$$P(Y|X) = \frac{P(X|Y) * P(Y)}{P(X)}$$

ignore P(X), since it's a constant

Y: Sentiment = neg or pos

*X: each word in T (predictable, with, no, fun)
 and we assume each word occurs
 independently*

Text	Sentiment
just plain boring	neg
entirely predictable and lacks energy	neg
no surprises and very few laughs	neg
very powerful	pos
the most fun film of the summer	pos

Predict the sentiment of a text T:
“predictable with no fun”

$$P(Y|X) = P(X|Y) * P(Y)$$

Prior probability

*Y: Sentiment = **pos***

*X: each word in T (predictable, with, no, fun)
 and we assume each word occurs
 independently*

$$P(Y) = P(pos) = 2/5$$

Text	Sentiment
just plain boring	neg
entirely predictable and lacks energy	neg
no surprises and very few laughs	neg
very powerful	pos
the most fun film of the summer	pos

Predict the sentiment of a text T:
“predictable with no fun”

$$P(Y|X) = P(X|Y) * 0.4$$

*Y: Sentiment = **pos***

*X: each word in T (predictable, with, no, fun)
 and we assume each word occurs
 independently*

$P(X|Y)$ with Laplace smoothing applied

$$= P(\text{word}_{\text{predictable}} | \text{pos}) * P(\text{word}_{\text{with}} | \text{pos}) * P(\text{word}_{\text{no}} | \text{pos}) * P(\text{word}_{\text{fun}} | \text{pos})$$

$$= \frac{0+1}{9+20} * \frac{0+1}{9+20} * \frac{1+1}{9+20} = \frac{1*1*2}{29^3}$$

**ignore this, since with doesn't exist
 in the training dataset**

Text	Sentiment
just plain boring	neg
entirely predictable and lacks energy	neg
no surprises and very few laughs	neg
very powerful	pos
the most fun film of the summer	pos

Predict the sentiment of a text T:
“predictable with no fun”

$$P(Y|X) = P(X|Y) * P(Y)$$

$$P(Y = pos | X) = \frac{0 + 1}{9 + 20} * \frac{0 + 1}{9 + 20} * \frac{1 + 1}{9 + 20} * 0.4 = 3.2 * 10^{-5}$$

$$P(Y = neg | X) = \frac{1 + 1}{14 + 20} * \frac{1 + 1}{14 + 20} * \frac{0 + 1}{14 + 20} * 0.6 = 6.1 * 10^{-5}$$

Naive Bayes Classifier Implementation

Lab 8 asks you to build a Naive-Bayes Classifier from the very scratch

Why?

It forces you to understand how the entire process works.. down to the nitty-gritty details.

Step I: Split texts in the training dataset into positive and negative groups

Text	Sentiment
just plain boring	neg
entirely predictable and lacks energy	neg
no surprises and very few laughs	neg
very powerful	pos
the most fun film of the summer	pos



Negative Only

Text
just plain boring
entirely predictable and lacks energy
no surprises and very few laughs

Positive Only

Text
very powerful
the most fun film of the summer

Step2: Calculate Prior probability

Negative Only

Text
just plain boring
entirely predictable and lacks energy
no surprises and very few laughs



Prior Probability

pos: 2/5

neg: 3/5

Positive Only

Text
very powerful
the most fun film of the summer



Step3: Count occurrences for each word, for each sentiment class

Negative Only

Text

just plain boring

entirely predictable and
lacks energy

no surprises and very
few laughs



just	1
plain	1
boring	1
entirely	1
predictable	1
and	2
lacks	1
energy	1
no	1
surprises	1
very	1
few	1
laughs	1

called “bag of words”
notation

Positive Only

Text

very powerful

the most fun film of the
summer

Step3: Count occurrences for each word, for each sentiment class

Negative Only

Text
just plain boring
entirely predictable and lacks energy
no surprises and very few laughs

Positive Only

Text
very powerful
the most fun film of the summer



very	1
powerful	1
the	1
most	1
fun	1
film	1
of	1
the	1
summer	1

For Step3, you need to break down each review text into words

1. tokenizing

- a. convert a text “just plain boring” to a list of words ['just', 'plain', 'boring']
- b. you will need to do some pre-processing

2. remove symbols

- a. remove extra spaces
- b. remove some strange characters

3. And more! (extra)

For Step3, you also need to count the number of total unique words

across pos/neg class, so that you know what number to add for the Laplace smoothing

Negative
Only

Text
just plain boring
entirely predictable and lacks energy
no surprises and very few laughs

Positive
Only

Text
very powerful
the most fun film of the summer

There are total 20 unique words:

just, plain, boring, entirely, predictable, and, lacks, energy, no, surprises, very, few, laughs, powerful, the, most, fun, film, of, summer

Step4: Calculate posterior probability, for each class, in a log form

I. the posterior probability is often very small

a. use `math.log()`, so that the program can handle small numbers

$$P(Y|X) = P(X|Y) * P(Y)$$

$$P(Y = pos | X) = \frac{0+1}{9+20} * \frac{0+1}{9+20} * \frac{1+1}{9+20} * 0.4 = 3.2 * 10^{-5}$$



You can get this information by looking at bag-of-words

$$\log(P(Y = pos | X)) = \log\left(\frac{0+1}{9+20}\right) + \log\left(\frac{0+1}{9+20}\right) + \log\left(\frac{1+1}{9+20}\right) + \log(0.4)$$

$$= \log(3.2 * 10^{-5})$$

Step5: Choose max of the two posterior probabilities

- I. if $\log(P(Y = pos|X))$ is larger than $\log(P(Y = neg|X))$, the text is likely to have positive sentiment; otherwise negative